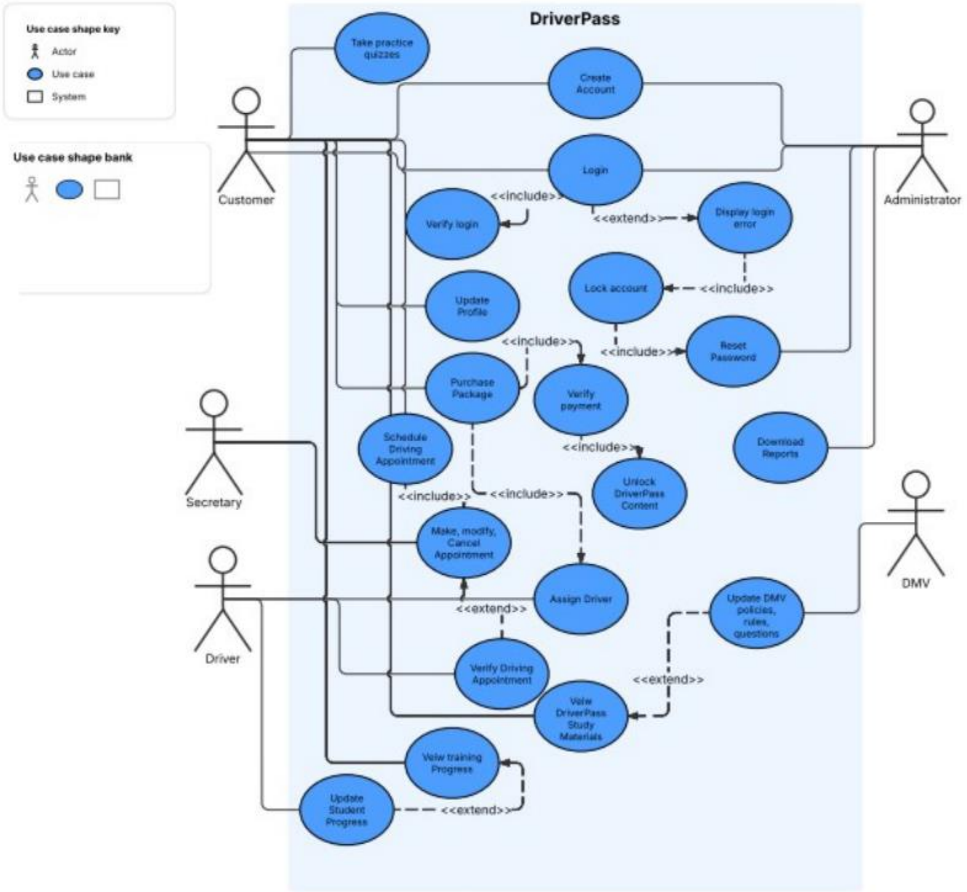
CS 255 System Design Document Template

This template lays out all the different sections that you need to complete for Project Two. Each section has guidance to prompt your thinking. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead the goal is to complete each section based on what your client's needs are. Remove this note when you are finished, and replace all bracketed text with the relevant information.

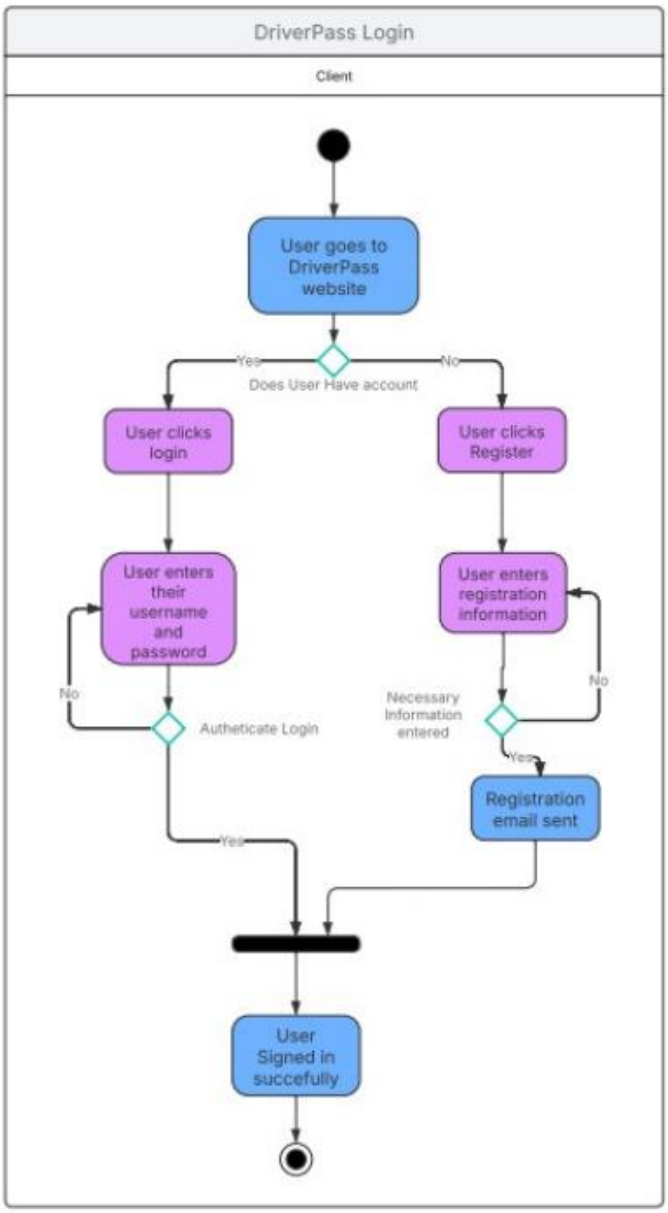
UML Diagrams

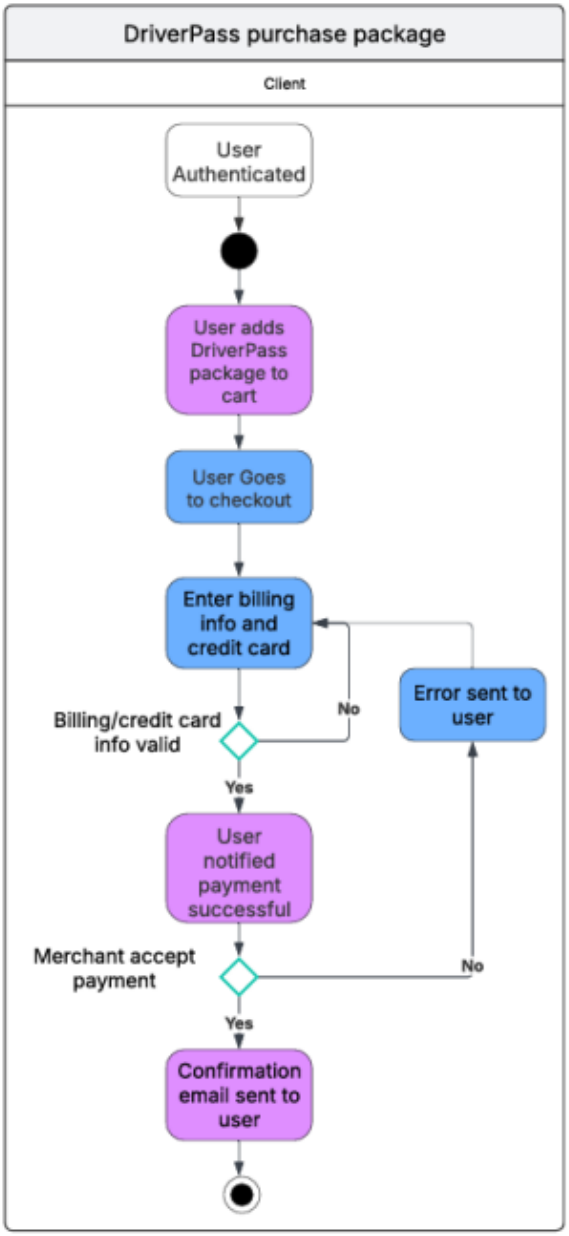
UML Use Case Diagram



UML Activity Diagrams

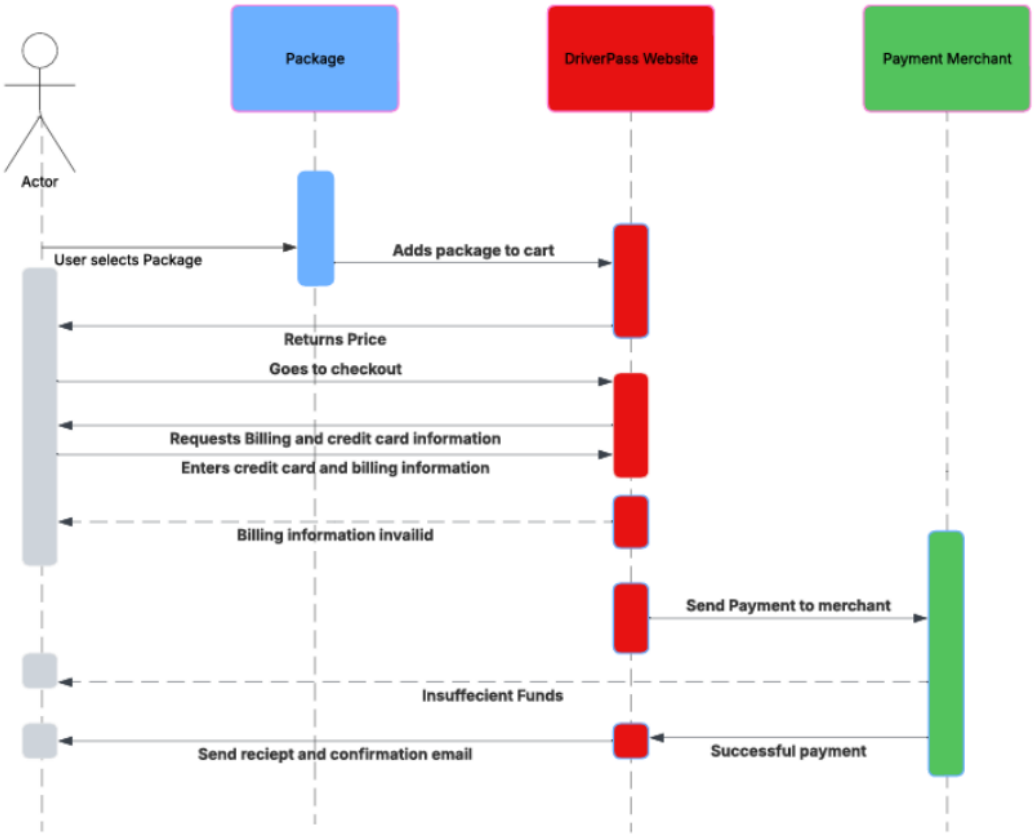
*[You were asked to choose **two** use cases and create **two** activity diagrams, one for each use case. Please insert **both** of your activity diagrams here. Check to make sure that you included appropriate components and symbols and that your design meets the client's needs.]*


DriverPass Login
Client
User goes to DriverPass website
Yes
No
Does User Have account
User clicks login
User clicks Register
User enters their username and password
User enters registration information
No
No
Autheticate Login
Necessary Information entered
Yes
Registration email sent
Yes
User Signed in succefully



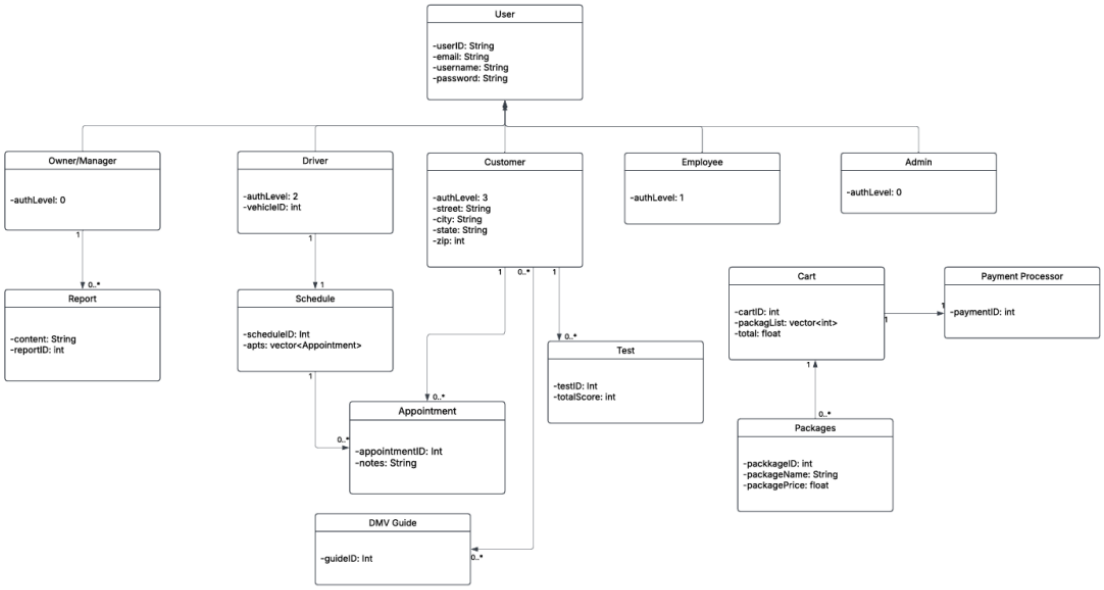
UML Sequence Diagram

*[You were asked to create a sequence diagram based on **one** of the use cases you chose. Please insert your sequence diagram here. Check to make sure that you included appropriate components and symbols and that your design meets the client's needs.]*



UML Class Diagram

*[You were asked to create a class diagram based on the different classes and attributes needed for your system design. You are **not** required to include methods, but you may if you wish. Please insert your class diagram here. Check to make sure that you included appropriate components and symbols and that your design meets the client's requirements.]*



Technical Requirements

[Based on the diagrams you have created, describe the technical requirements of your system. These requirements should address the required hardware, software, tools, and infrastructure necessary for your system design.]

A suitable technical implementation would require a cloud-based server to host all student, teacher, lesson, package, and staff information in a robust and scalable platform. I would suggest the use of MySQL as the primary database management system. The system could be developed using a JavaScript front end with either Java or Python being suitable for the back end, with my preference being Python. I would recommend the use of Visual Studio Code as the main Integrated Development Environment (IDE) along with Git for version control. Hosting the system in the cloud provides the benefits of increased storage and scaling of resources as required. In addition to these benefits, it is also essential that the system has a robust backup policy to prevent data loss in the event of hardware or software failure.